1. Product Info

The BM-3221 / BK3221 Bluetooth-Compatible Audio Receiver Module is a high-performance, cost-effective wireless audio solution engineered specifically for DIY smart speakers, customized car audio modifications, wireless headsets, and generic IoT audio streaming integrations. Driven by the highly stable Broadcom BK3221 core architecture chip, this ultra-compact surface-mount module integrates a complete radio frequency transceiver, an audio decoder engine, a high-accuracy T260 crystal oscillator, and an external SPI flash memory block on a single streamlined blue substrate layout. Native support for Advanced Audio Distribution Profile (A2DP), Audio/Video Remote Control Profile (AVRCP), and Hands-Free Profile (HFP) allows the board to decode high-fidelity wireless audio streams directly from modern mobile smartphones, laptops, and tablets. Equipped with an onboard high-efficiency serpentine PCB trace antenna and gold-plated castellated edge pads, it provides engineering researchers and audio hardware developers with a premium, plug-and-play brick for wireless sound transmission projects.

1. Safety & Warnings

Strict Power Supply Boundaries: The nominal operating voltage for this Bluetooth audio module is DC 3.3V, with a strict functional safety window between 2.8V and 3.6V. Do not connect the VCC pad directly to standard 5V microcontroller lines or un-regulated battery cells, as over-voltage surges past the 3.6V ceiling will instantly cause permanent thermal breakdown of the Broadcom microchip.

Electrostatic Sensitivity Notice: This wireless audio patch contains high-frequency integrated microwave logic gates and sensitive CMOS registers. It is highly vulnerable to Electrostatic Discharge (ESD). Assembly technicians must wear grounded anti-static wrist straps or handle the module exclusively by its insulating edges on an ESD-safe mat to prevent silent gate punctures.

Antenna Radiation Keep-Out Zone: The integrated golden serpentine PCB antenna section must remain entirely free of mechanical obstructions. Do not route heavy copper tracks, power ground planes, battery wires, or metallic chassis fasteners directly above, underneath, or adjacent to the antenna window loop. Placing metal shields in this keep-out area will absorb the 2.4GHz radio waves, drastically shortening the transmission range or causing severe audio stuttering.

Thermal Solder Joint Protection: During surface-mount component reflow or manual hand-soldering onto prototype carrier boards, strictly regulate your soldering iron tip temperature and keep pad contact time below 3 seconds. Prolonged high-heat exposure will lift the fine castellated copper rings or crack the adjacent crystal resonator housing.

1. Instructions for Use

Hardware Alignment and Assembly: Place the module flat onto the corresponding copper layout pattern of your target breakout adapter or custom carrier motherboard. Align the castellated edge pads carefully, then use a fine electronics soldering tip and high-quality resin-core solder to establish stable, low-resistance connections along the terminal rows.

Power and Peripheral Connections: Connect the VCC pad to a clean, regulated DC 3.3V power

source line and tie the GND pad directly to your main circuit common ground plane. Wire the analog audio output lines (Left channel, Right channel, and Audio Common Ground) straight to the input terminals of an external audio power amplifier module or headphones driver stage. Do not connect passive speakers directly to the low-power chip outputs.

Implementing Control Keys and Status Indicators: Solder standard micro-tactile switches across the designated GPIO breakout pads (such as Play/Pause, Volume Up/Next Track, Volume Down/Previous Track) to enable manual control loops. Connect a low-current status LED through a suitable inline limiting resistor to the status output pin to track wireless pairing states (e.g., rapid flashing for pairing mode, solid lit for an active link).

System Pairing and Software Flashing: Power up the entire circuit framework. Open the Bluetooth settings menu on your host smartphone or transmission source device, search for the module's broadcast identifier, and initiate pairing. The embedded AVRCP and A2DP profile layers will automatically handle wireless synchronization, allowing you to stream high-fidelity digital music streams or operate hands-free call audio loops instantly.

1. MAINTENANCE & STORAGE

Flux Residue Cleanliness: Upon completing your manual soldering routine, thoroughly clean the gaps between the fine castellated pins using a small stiff-bristled brush and pure electronics-grade isopropyl alcohol to remove brown rosin patches and guard against moisture tracking.

Check Antenna Path: Periodically inspect the module to ensure no ambient moisture films, airborne dust blankets, or loose metallic shavings have settled onto the PCB trace antenna section, which could dampen RF sensitivity thresholds.

Sealed Storage Environment: Keep uninstalled wireless audio modules packed inside individual anti-static shielding pouches containing fresh desiccant packs to prevent pad oxidation. Store the components in a cool, clean, and dry warehouse area with storage temperatures regulated between -40 and +85 degrees Celsius.

1. DISPOSAL & RECYCLING

Technical Scrap Categorization: This high-frequency Bluetooth audio transceiver module represents advanced electronic scrap, comprising a multi-layer fiberglass carrier board substrate, an integrated silicon CMOS logic controller chip, an external surface-mount flash memory microchip, a metallic crystal container, and gold-plated connection trails.

Eco-Safe Material Reclamation: Do not discard this wireless hardware module alongside regular domestic household garbage bins or municipal rubbish sorting channels. Please transport unserviceable modules to an officially certified local electronic waste processing depot to support safe raw material separation and guard environmental ecosystems.

1. TECHNICAL SPECIFICATIONS

Component Type: High-Performance Bluetooth-Compatible Wireless Audio Receiver Module

Hardware Model Designation: BM-3221 / BK3221 SMT Series

Core Processor Architecture Chipset: Broadcom BK3221 Audio SoC Platform

Operational Frequency Spectrum: Standard 2.4GHz ISM Wireless Band

Integrated Wireless Protocol Profiles: Advanced Audio Distribution Profile (A2DP), Audio/Video Remote Control Profile (AVRCP), and Hands-Free Profile (HFP)

Digital Modulation Formats Support: GFSK, PI/4 DQPSK, and 8DPSK radio frequency tracking

Onboard Frequency Reference Source: High-stability T260 quartz crystal oscillator engine

Integrated Storage Component: Onboard dedicated high-speed SPI Flash memory chip

Core Operating Input Voltage Range (VCC): DC 2.8V to 3.6V (DC 3.3V typical operational rail)

Wireless Signal Transmission Power: Standard Class 2 output levels supporting a stable open-air line-of-sight range up to 10 meters

Receiver RF Sensitivity Rating: -80 dBm typical threshold baseline for stable wireless synchronization

Audio Processing Performance: High-fidelity low-noise analog stereo sound output

Interface Interconnect Footprint: Golden castellated multi-pin edge breakout tracks for surface-mount design integration

Onboard Radiator System: Integrated 2.4GHz serpentine layout PCB copper trace antenna

Primary Board Substrate Construction: High-grade double-sided flame-retardant fiberglass PCB

Safe System Operating Temperature Scope: -20 to +70 degrees Celsius

Safe System Storage Temperature Spectrum: -40 to +85 degrees Celsius